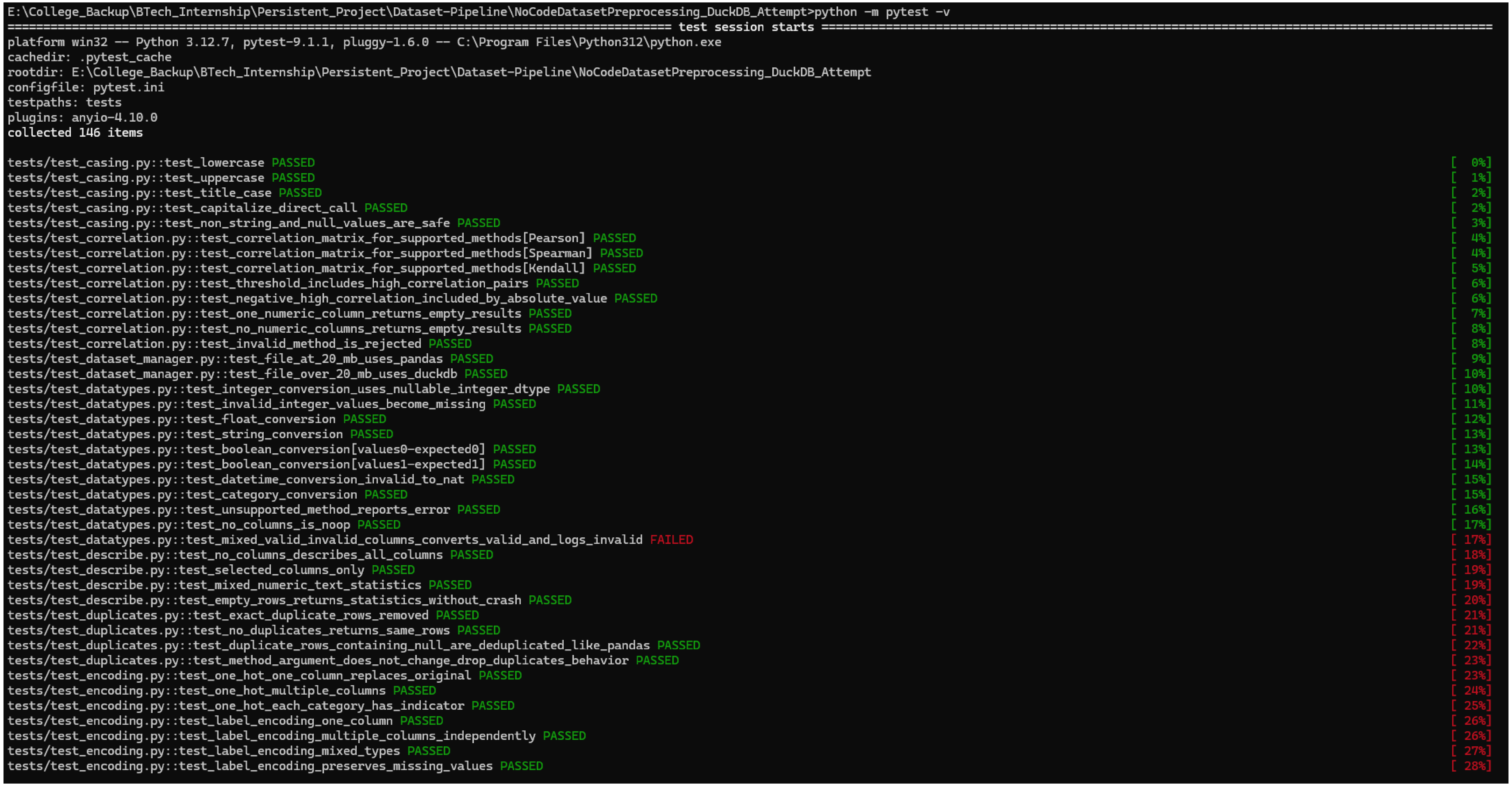
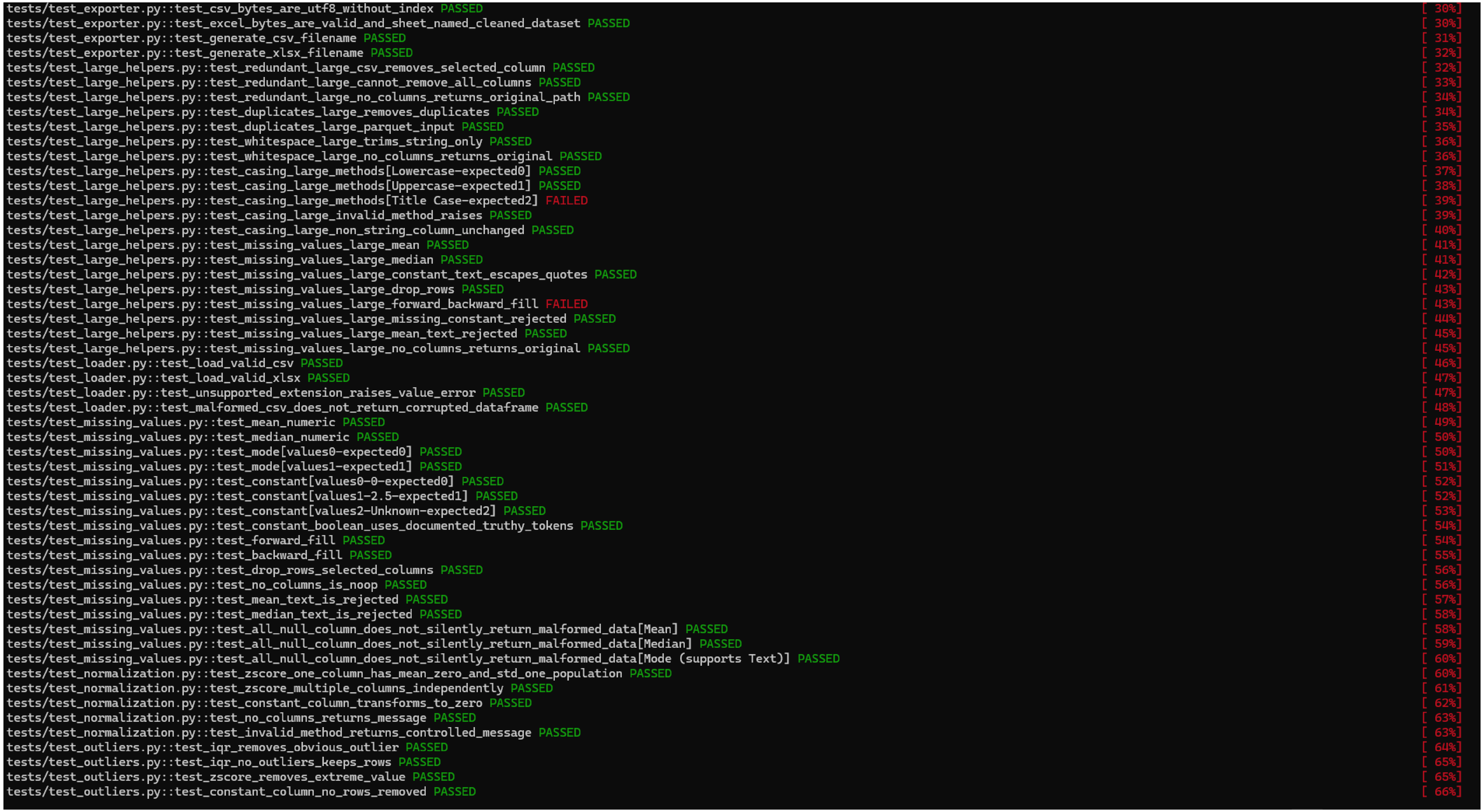
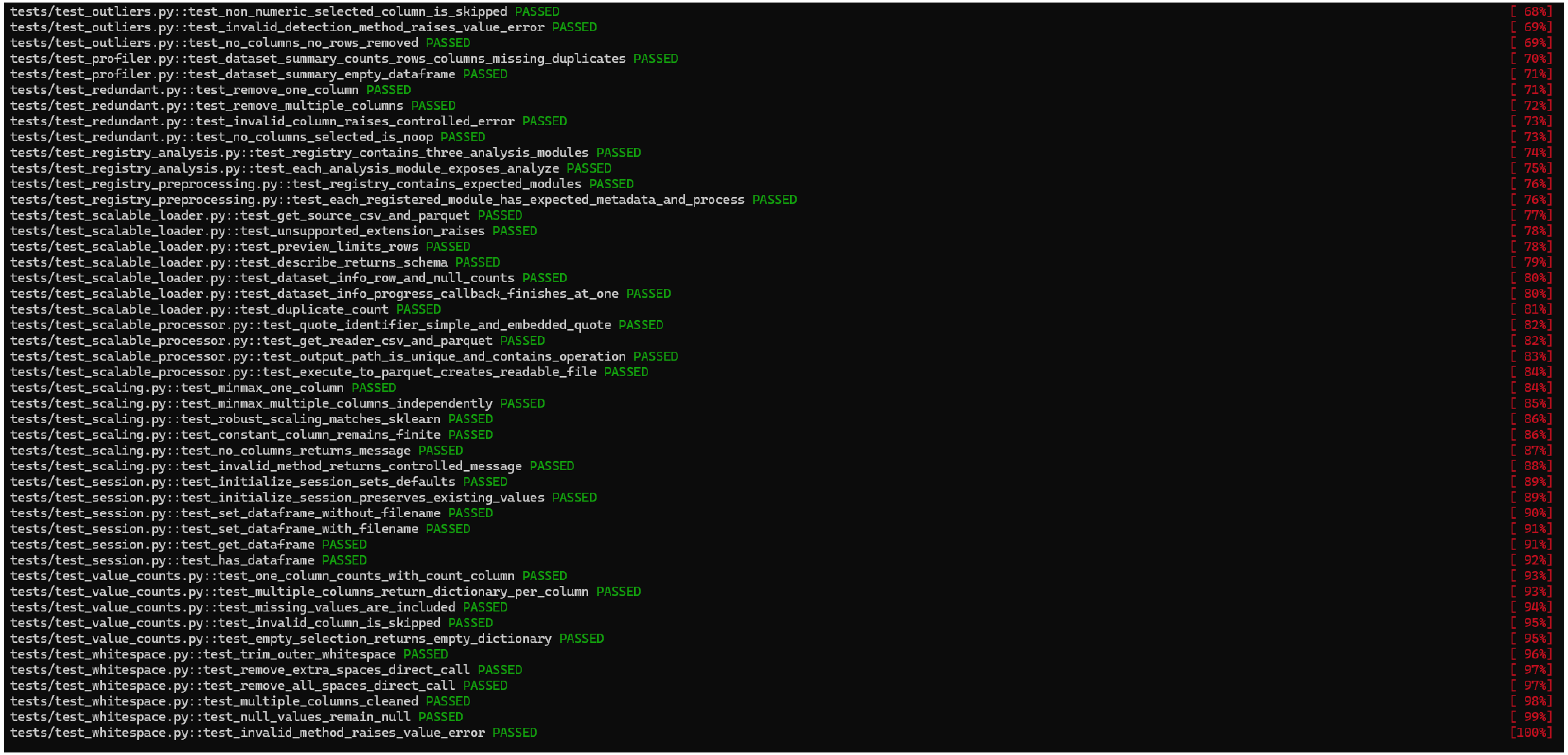


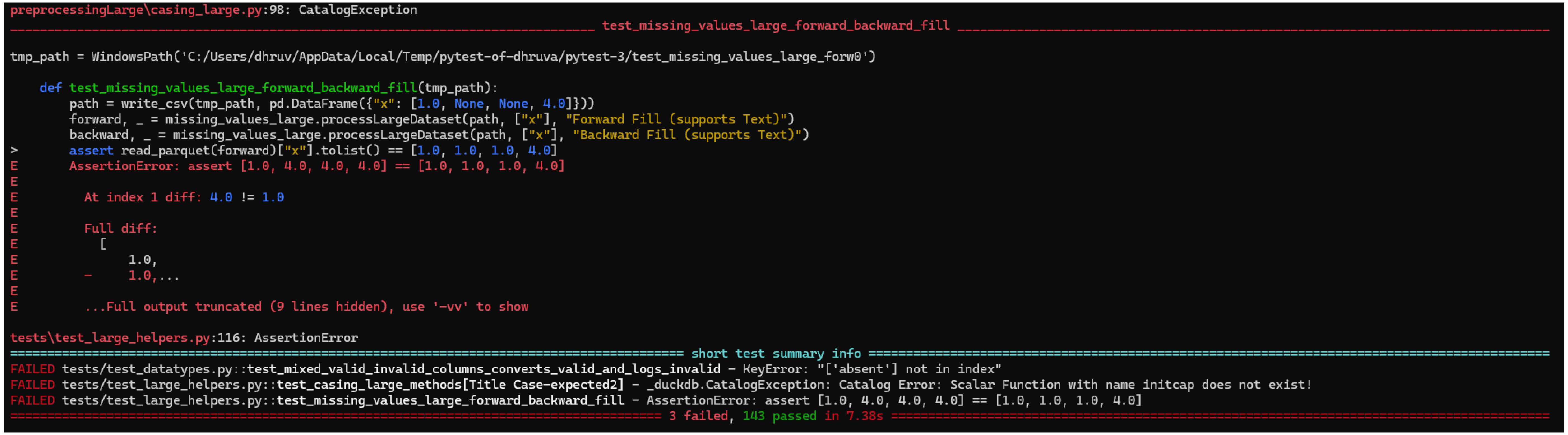
```
E:\College_Backup\BTech_Internship\Persistent_Project\Dataset-Pipeline\NoCodeDatasetPreprocessing_DuckDB_Attempt>python -m pytest -v
============================================================================================= test session starts ==============================================================================================
platform win32 -- Python 3.12.7, pytest-9.1.1, pluggy-1.6.0 -- C:\Program Files\Python312\python.exe
cachedir: .pytest_cache
rootdir: E:\College_Backup\BTech_Internship\Persistent_Project\Dataset-Pipeline\NoCodeDatasetPreprocessing_DuckDB_Attempt
configfile: pytest.ini
testpaths: tests
plugins: anyio-4.10.0
collected 146 items

tests/test_casing.py::test_lowercase PASSED                                                                                                                                                               [  0%]
tests/test_casing.py::test_uppercase PASSED                                                                                                                                                               [  1%]
tests/test_casing.py::test_title_case PASSED                                                                                                                                                              [  2%]
tests/test_casing.py::test_capitalize_direct_call PASSED                                                                                                                                                  [  2%]
tests/test_casing.py::test_non_string_and_null_values_are_safe PASSED                                                                                                                                     [  3%]
tests/test_correlation.py::test_correlation_matrix_for_supported_methods[Pearson] PASSED                                                                                                                  [  4%]
tests/test_correlation.py::test_correlation_matrix_for_supported_methods[Spearman] PASSED                                                                                                                 [  4%]
tests/test_correlation.py::test_correlation_matrix_for_supported_methods[Kendall] PASSED                                                                                                                  [  5%]
tests/test_correlation.py::test_threshold_includes_high_correlation_pairs PASSED                                                                                                                          [  6%]
tests/test_correlation.py::test_negative_high_correlation_included_by_absolute_value PASSED                                                                                                               [  6%]
tests/test_correlation.py::test_one_numeric_column_returns_empty_results PASSED                                                                                                                           [  7%]
tests/test_correlation.py::test_no_numeric_columns_returns_empty_results PASSED                                                                                                                           [  8%]
tests/test_correlation.py::test_invalid_method_is_rejected PASSED                                                                                                                                         [  8%]
tests/test_dataset_manager.py::test_file_at_20_mb_uses_pandas PASSED                                                                                                                                      [  9%]
tests/test_dataset_manager.py::test_file_over_20_mb_uses_duckdb PASSED                                                                                                                                    [ 10%]
tests/test_datatypes.py::test_integer_conversion_uses_nullable_integer_dtype PASSED                                                                                                                       [ 10%]
tests/test_datatypes.py::test_invalid_integer_values_become_missing PASSED                                                                                                                                [ 11%]
tests/test_datatypes.py::test_float_conversion PASSED                                                                                                                                                     [ 12%]
tests/test_datatypes.py::test_string_conversion PASSED                                                                                                                                                    [ 13%]
tests/test_datatypes.py::test_boolean_conversion[values0-expected0] PASSED                                                                                                                                [ 13%]
tests/test_datatypes.py::test_boolean_conversion[values1-expected1] PASSED                                                                                                                                [ 14%]
tests/test_datatypes.py::test_datetime_conversion_invalid_to_nat PASSED                                                                                                                                   [ 15%]
tests/test_datatypes.py::test_category_conversion PASSED                                                                                                                                                  [ 15%]
tests/test_datatypes.py::test_unsupported_method_reports_error PASSED                                                                                                                                     [ 16%]
tests/test_datatypes.py::test_no_columns_is_noop PASSED                                                                                                                                                   [ 17%]
tests/test_datatypes.py::test_mixed_valid_invalid_columns_converts_valid_and_logs_invalid FAILED                                                                                                          [ 17%]
tests/test_describe.py::test_no_columns_describes_all_columns PASSED                                                                                                                                      [ 18%]
tests/test_describe.py::test_selected_columns_only PASSED                                                                                                                                                 [ 19%]
tests/test_describe.py::test_mixed_numeric_text_statistics PASSED                                                                                                                                         [ 19%]
tests/test_describe.py::test_empty_rows_returns_statistics_without_crash PASSED                                                                                                                           [ 20%]
tests/test_duplicates.py::test_exact_duplicate_rows_removed PASSED                                                                                                                                        [ 21%]
tests/test_duplicates.py::test_no_duplicates_returns_same_rows PASSED                                                                                                                                     [ 21%]
tests/test_duplicates.py::test_duplicate_rows_containing_null_are_deduplicated_like_pandas PASSED                                                                                                         [ 22%]
tests/test_duplicates.py::test_method_argument_does_not_change_drop_duplicates_behavior PASSED                                                                                                            [ 23%]
tests/test_encoding.py::test_one_hot_one_column_replaces_original PASSED                                                                                                                                  [ 23%]
tests/test_encoding.py::test_one_hot_multiple_columns PASSED                                                                                                                                              [ 24%]
tests/test_encoding.py::test_one_hot_each_category_has_indicator PASSED                                                                                                                                   [ 25%]
tests/test_encoding.py::test_label_encoding_one_column PASSED                                                                                                                                             [ 26%]
tests/test_encoding.py::test_label_encoding_multiple_columns_independently PASSED                                                                                                                         [ 26%]
tests/test_encoding.py::test_label_encoding_mixed_types PASSED                                                                                                                                            [ 27%]
tests/test_encoding.py::test_label_encoding_preserves_missing_values PASSED                                                                                                                               [ 28%]
```



```
tests/test_exporter.py::test_csv_bytes_are_utf8_without_index PASSED                                                                                                                                      [ 30%]
tests/test_exporter.py::test_excel_bytes_are_valid_and_sheet_named_cleaned_dataset PASSED                                                                                                                 [ 30%]
tests/test_exporter.py::test_generate_csv_filename PASSED                                                                                                                                                 [ 31%]
tests/test_exporter.py::test_generate_xlsx_filename PASSED                                                                                                                                                [ 32%]
tests/test_large_helpers.py::test_redundant_large_csv_removes_selected_column PASSED                                                                                                                      [ 32%]
tests/test_large_helpers.py::test_redundant_large_cannot_remove_all_columns PASSED                                                                                                                        [ 33%]
tests/test_large_helpers.py::test_redundant_large_no_columns_returns_original_path PASSED                                                                                                                 [ 34%]
tests/test_large_helpers.py::test_duplicates_large_removes_duplicates PASSED                                                                                                                              [ 34%]
tests/test_large_helpers.py::test_duplicates_large_parquet_input PASSED                                                                                                                                   [ 35%]
tests/test_large_helpers.py::test_whitespace_large_trims_string_only PASSED                                                                                                                               [ 36%]
tests/test_large_helpers.py::test_whitespace_large_no_columns_returns_original PASSED                                                                                                                     [ 36%]
tests/test_large_helpers.py::test_casing_large_methods[Lowercase-expected0] PASSED                                                                                                                        [ 37%]
tests/test_large_helpers.py::test_casing_large_methods[Uppercase-expected1] PASSED                                                                                                                        [ 38%]
tests/test_large_helpers.py::test_casing_large_methods[Title Case-expected2] FAILED                                                                                                                       [ 39%]
tests/test_large_helpers.py::test_casing_large_invalid_method_raises PASSED                                                                                                                               [ 39%]
tests/test_large_helpers.py::test_casing_large_non_string_column_unchanged PASSED                                                                                                                         [ 40%]
tests/test_large_helpers.py::test_missing_values_large_mean PASSED                                                                                                                                        [ 41%]
tests/test_large_helpers.py::test_missing_values_large_median PASSED                                                                                                                                      [ 41%]
tests/test_large_helpers.py::test_missing_values_large_constant_text_escapes_quotes PASSED                                                                                                                [ 42%]
tests/test_large_helpers.py::test_missing_values_large_drop_rows PASSED                                                                                                                                   [ 43%]
tests/test_large_helpers.py::test_missing_values_large_forward_backward_fill FAILED                                                                                                                       [ 43%]
tests/test_large_helpers.py::test_missing_values_large_missing_constant_rejected PASSED                                                                                                                   [ 44%]
tests/test_large_helpers.py::test_missing_values_large_mean_text_rejected PASSED                                                                                                                          [ 45%]
tests/test_large_helpers.py::test_missing_values_large_no_columns_returns_original PASSED                                                                                                                 [ 45%]
tests/test_loader.py::test_load_valid_csv PASSED                                                                                                                                                          [ 46%]
tests/test_loader.py::test_load_valid_xlsx PASSED                                                                                                                                                         [ 47%]
tests/test_loader.py::test_unsupported_extension_raises_value_error PASSED                                                                                                                                [ 47%]
tests/test_loader.py::test_malformed_csv_does_not_return_corrupted_dataframe PASSED                                                                                                                       [ 48%]
tests/test_missing_values.py::test_mean_numeric PASSED                                                                                                                                                    [ 49%]
tests/test_missing_values.py::test_median_numeric PASSED                                                                                                                                                  [ 50%]
tests/test_missing_values.py::test_mode[values0-expected0] PASSED                                                                                                                                         [ 50%]
tests/test_missing_values.py::test_mode[values1-expected1] PASSED                                                                                                                                         [ 51%]
tests/test_missing_values.py::test_constant[values0-0-expected0] PASSED                                                                                                                                   [ 52%]
tests/test_missing_values.py::test_constant[values1-2.5-expected1] PASSED                                                                                                                                 [ 52%]
tests/test_missing_values.py::test_constant[values2-Unknown-expected2] PASSED                                                                                                                             [ 53%]
tests/test_missing_values.py::test_constant_boolean_uses_documented_truthy_tokens PASSED                                                                                                                  [ 54%]
tests/test_missing_values.py::test_forward_fill PASSED                                                                                                                                                    [ 54%]
tests/test_missing_values.py::test_backward_fill PASSED                                                                                                                                                   [ 55%]
tests/test_missing_values.py::test_drop_rows_selected_columns PASSED                                                                                                                                      [ 56%]
tests/test_missing_values.py::test_no_columns_is_noop PASSED                                                                                                                                              [ 56%]
tests/test_missing_values.py::test_mean_text_is_rejected PASSED                                                                                                                                           [ 57%]
tests/test_missing_values.py::test_median_text_is_rejected PASSED                                                                                                                                         [ 58%]
tests/test_missing_values.py::test_all_null_column_does_not_silently_return_malformed_data[Mean] PASSED                                                                                                   [ 58%]
tests/test_missing_values.py::test_all_null_column_does_not_silently_return_malformed_data[Median] PASSED                                                                                                 [ 59%]
tests/test_missing_values.py::test_all_null_column_does_not_silently_return_malformed_data[Mode (supports Text)] PASSED                                                                                   [ 60%]
tests/test_normalization.py::test_zscore_one_column_has_mean_zero_and_std_one_population PASSED                                                                                                           [ 60%]
tests/test_normalization.py::test_zscore_multiple_columns_independently PASSED                                                                                                                            [ 61%]
tests/test_normalization.py::test_constant_column_transforms_to_zero PASSED                                                                                                                               [ 62%]
tests/test_normalization.py::test_no_columns_returns_message PASSED                                                                                                                                       [ 63%]
tests/test_normalization.py::test_invalid_method_returns_controlled_message PASSED                                                                                                                        [ 63%]
tests/test_outliers.py::test_iqr_removes_obvious_outlier PASSED                                                                                                                                           [ 64%]
tests/test_outliers.py::test_iqr_no_outliers_keeps_rows PASSED                                                                                                                                            [ 65%]
tests/test_outliers.py::test_zscore_removes_extreme_value PASSED                                                                                                                                          [ 65%]
tests/test_outliers.py::test_constant_column_no_rows_removed PASSED                                                                                                                                       [ 66%]
```



```
tests/test_outliers.py::test_non_numeric_selected_column_is_skipped PASSED                                    [ 68%]
tests/test_outliers.py::test_invalid_detection_method_raises_value_error PASSED                               [ 69%]
tests/test_outliers.py::test_no_columns_no_rows_removed PASSED                                                [ 69%]
tests/test_profiler.py::test_dataset_summary_counts_rows_columns_missing_duplicates PASSED                    [ 70%]
tests/test_profiler.py::test_dataset_summary_empty_dataframe PASSED                                           [ 71%]
tests/test_redundant.py::test_remove_one_column PASSED                                                        [ 71%]
tests/test_redundant.py::test_remove_multiple_columns PASSED                                                  [ 72%]
tests/test_redundant.py::test_invalid_column_raises_controlled_error PASSED                                   [ 73%]
tests/test_redundant.py::test_no_columns_selected_is_noop PASSED                                              [ 73%]
tests/test_registry_analysis.py::test_registry_contains_three_analysis_modules PASSED                         [ 74%]
tests/test_registry_analysis.py::test_each_analysis_module_exposes_analyze PASSED                             [ 75%]
tests/test_registry_preprocessing.py::test_registry_contains_expected_modules PASSED                          [ 76%]
tests/test_registry_preprocessing.py::test_each_registered_module_has_expected_metadata_and_process PASSED    [ 76%]
tests/test_scalable_loader.py::test_get_source_csv_and_parquet PASSED                                         [ 77%]
tests/test_scalable_loader.py::test_unsupported_extension_raises PASSED                                       [ 78%]
tests/test_scalable_loader.py::test_preview_limits_rows PASSED                                                [ 78%]
tests/test_scalable_loader.py::test_describe_returns_schema PASSED                                            [ 79%]
tests/test_scalable_loader.py::test_dataset_info_row_and_null_counts PASSED                                   [ 80%]
tests/test_scalable_loader.py::test_dataset_info_progress_callback_finishes_at_one PASSED                     [ 80%]
tests/test_scalable_loader.py::test_duplicate_count PASSED                                                    [ 81%]
tests/test_scalable_processor.py::test_quote_identifier_simple_and_embedded_quote PASSED                      [ 82%]
tests/test_scalable_processor.py::test_get_reader_csv_and_parquet PASSED                                      [ 82%]
tests/test_scalable_processor.py::test_output_path_is_unique_and_contains_operation PASSED                    [ 83%]
tests/test_scalable_processor.py::test_execute_to_parquet_creates_readable_file PASSED                        [ 84%]
tests/test_scaling.py::test_minmax_one_column PASSED                                                          [ 84%]
tests/test_scaling.py::test_minmax_multiple_columns_independently PASSED                                      [ 85%]
tests/test_scaling.py::test_robust_scaling_matches_sklearn PASSED                                             [ 86%]
tests/test_scaling.py::test_constant_column_remains_finite PASSED                                             [ 86%]
tests/test_scaling.py::test_no_columns_returns_message PASSED                                                 [ 87%]
tests/test_scaling.py::test_invalid_method_returns_controlled_message PASSED                                  [ 88%]
tests/test_session.py::test_initialize_session_sets_defaults PASSED                                           [ 89%]
tests/test_session.py::test_initialize_session_preserves_existing_values PASSED                               [ 89%]
tests/test_session.py::test_set_dataframe_without_filename PASSED                                             [ 90%]
tests/test_session.py::test_set_dataframe_with_filename PASSED                                                [ 91%]
tests/test_session.py::test_get_dataframe PASSED                                                              [ 91%]
tests/test_session.py::test_has_dataframe PASSED                                                              [ 92%]
tests/test_value_counts.py::test_one_column_counts_with_count_column PASSED                                   [ 93%]
tests/test_value_counts.py::test_multiple_columns_return_dictionary_per_column PASSED                         [ 93%]
tests/test_value_counts.py::test_missing_values_are_included PASSED                                           [ 94%]
tests/test_value_counts.py::test_invalid_column_is_skipped PASSED                                             [ 95%]
tests/test_value_counts.py::test_empty_selection_returns_empty_dictionary PASSED                              [ 95%]
tests/test_whitespace.py::test_trim_outer_whitespace PASSED                                                   [ 96%]
tests/test_whitespace.py::test_remove_extra_spaces_direct_call PASSED                                         [ 97%]
tests/test_whitespace.py::test_remove_all_spaces_direct_call PASSED                                           [ 97%]
tests/test_whitespace.py::test_multiple_columns_cleaned PASSED                                                [ 98%]
tests/test_whitespace.py::test_null_values_remain_null PASSED                                                 [ 99%]
tests/test_whitespace.py::test_invalid_method_raises_value_error PASSED                                       [100%]
```



```
preprocessingLarge\casing_large.py:98: CatalogException
_______________________________________________________________________________ test_missing_values_large_forward_backward_fill _______________________________________________________________________________

tmp_path = WindowsPath('C:/Users/dhruv/AppData/Local/Temp/pytest-of-dhruva/pytest-3/test_missing_values_large_forw0')

    def test_missing_values_large_forward_backward_fill(tmp_path):
        path = write_csv(tmp_path, pd.DataFrame({"x": [1.0, None, None, 4.0]}))
        forward, _ = missing_values_large.processLargeDataset(path, ["x"], "Forward Fill (supports Text)")
        backward, _ = missing_values_large.processLargeDataset(path, ["x"], "Backward Fill (supports Text)")
>       assert read_parquet(forward)["x"].tolist() == [1.0, 1.0, 1.0, 4.0]
E       AssertionError: assert [1.0, 4.0, 4.0, 4.0] == [1.0, 1.0, 1.0, 4.0]
E
E         At index 1 diff: 4.0 != 1.0
E
E         Full diff:
E           [
E               1.0,
E         -     1.0,...
E
E         ...Full output truncated (9 lines hidden), use '-vv' to show

tests\test_large_helpers.py:116: AssertionError
=========================================================================================== short test summary info ============================================================================================
FAILED tests/test_datatypes.py::test_mixed_valid_invalid_columns_converts_valid_and_logs_invalid - KeyError: "['absent'] not in index"
FAILED tests/test_large_helpers.py::test_casing_large_methods[Title Case-expected2] - _duckdb.CatalogException: Catalog Error: Scalar Function with name initcap does not exist!
FAILED tests/test_large_helpers.py::test_missing_values_large_forward_backward_fill - AssertionError: assert [1.0, 4.0, 4.0, 4.0] == [1.0, 1.0, 1.0, 4.0]
======================================================================================== 3 failed, 143 passed in 7.38s =========================================================================================
```